04-23 第2篇

阅读理解（说明文） Spider Silk: A Thread Stronger Than Steel

① When most people hear the word "**spider**", they think of a small, hairy visitor on a wall. But scientists who study spiders are often not looking at the spider itself. They are looking at the **thin thread** that comes out of its body — **spider silk**. Thin as it is, this little white thread has become one of the most exciting materials of the 21st century.

② Spider silk is **incredibly strong**. Gram for gram, it is **five times stronger than steel** and still stays soft. If you took a long single thread and made it as thick as your finger, scientists say it could stop a passenger jet in the air. No man-made material has yet reached this mix of **strength** and **softness** in one package. Metals are strong but break easily when bent; rubber stretches but pulls apart. Spider silk has both.

③ How does a spider make it? Deep inside the spider's body, there are small body parts called **silk glands**. They store a thick **liquid protein**. When the spider is ready, it pushes this liquid through tiny holes on the back end of its body. The liquid changes into solid thread in less than a second, the way a hot candle hardens in cold air. Different glands can also make different kinds of silk — one kind for the **sticky** net that catches flies, and another kind for the **strong frame** that holds the whole web together.

④ Spiders use silk in clever ways. Some tiny spiders climb to a high leaf, release a long thread, and let the wind **carry them through the sky** like a small paper plane. Baby spiders use silk to **travel up to hundreds of kilometres**. Other spiders use silk to build a **small underwater bell**, filled with air, where they can rest deep inside a pond. Each spider, in a way, is a tiny factory of many products.

⑤ Because spider silk is so useful, people have tried to use it for a long time. In the 19th century, some European soldiers used it in rifle-sights. Doctors have used spider silk to help **heal wounds**. Today scientists are trying to make **artificial spider silk**: they put the spider's silk gene into **goats, bacteria, or silkworms**, so that these animals can produce the protein. From that protein, factories hope to make light bullet-stop vests, strong ropes, even thin medical **stitches** that disappear inside our skin.

⑥ Of course, not every project is a success yet. Real spider silk is still very hard to copy in large amounts. Spiders cannot be farmed like silkworms — they prefer to live alone, and they may eat each other in a shared box. So scientists keep trying new paths. In the future, the clothes on your back, the rope in your gym, and the stitches on your doctor's table may all quietly thank a small eight-legged teacher: the spider.

根据文章内容，选择最佳答案。

1. What is the main topic of the article?

A. How to kill spiders in the house.
B. Why spider silk is amazing and how people try to use it.
C. How to feed pet spiders.
D. The history of cotton.

2. Compared with steel, spider silk is:

A. Weaker but softer.
B. Gram for gram about 5 times stronger and still soft.
C. Heavier and harder.
D. The same in every way.

3. How is spider silk made inside the spider's body?

A. It is born hard and sharp.
B. A thick liquid protein in silk glands is pushed out and hardens into solid thread.
C. It is built from dust and water.
D. It grows like a leaf.

4. Which use of silk is NOT mentioned in the article?

A. Tiny spiders flying through the sky with a long thread.
B. An underwater air-filled bell for resting.
C. Healing wounds for people.
D. Cooking silk with rice for dinner.

5. Why has real spider silk been hard to produce in large amounts?

A. Because spiders live alone and may attack each other in a shared box.
B. Because spiders are too big.
C. Because silk is illegal.
D. Because spiders live in very cold caves only.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】第①段末 + 全文
【解析】讲蜘蛛丝的神奇与应用。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "5 times stronger than steel"
【解析】每克强度为钢的五倍。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第③段 silk glands 描述
【解析】液态蛋白从腺体挤出后迅速硬化。故选 B。

4.
【答案】D
【题型】细节理解题
【定位】第④—⑤段用途清单
【解析】没有提到"煮饭食用"。故选 D。

5.
【答案】A
【题型】细节理解题
【定位】第⑥段 "prefer to live alone... eat each other"
【解析】蜘蛛无法像蚕一样群养。故选 A。

词汇注释（超纲词）

- **spider silk** [n.] 蜘蛛丝
- **gland** [n.] 腺体
- **stitch** [n.] 缝合线/针脚